Vivek Jude Manoj

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Education

University of Waterloo — Waterloo, Ontario
Bachelor of Computer Science, Artificial Intelligence Specialization — *Expected Graduation: December 2027*

- Coursework: Distributed Systems, Operating Systems, Networks, Databases, Security, Machine Learning

Technical Skills

Languages: Go, Python, Java, C/C++, TypeScript/JavaScript, Rust, SQL
Infrastructure & Cloud: AWS, Kubernetes, Docker, Terraform, Bazel, Linux, Jenkins, GitHub Actions
Backend & Observability: gRPC/Protobuf, REST, GraphQL, PostgreSQL, Redis, OpenTelemetry, Prometheus
AI/ML: LLM Agents (MCP), Gemini API, PyTorch, TensorFlow, BERT, scikit-learn

Experience

Shopify — Jan 2027 – Apr 2027
Software Engineer Intern (Incoming), Backend & Infrastructure — *Toronto, Ontario*

Arctic Wolf Networks — Jan 2026 – Apr 2026
Software Engineer Intern, Infrastructure & Developer Productivity — *Waterloo, Ontario*

- Shipped an **AWS Dedicated Host Allocator** in **Go** and **Python**; adopted by **3+ teams** to automate EC2 provisioning, saving developers ∼**5 hours** per allocation and reducing accidental cloud spend.
- Provisioned **5+ macOS CI build nodes** on AWS EC2 Dedicated Hosts end-to-end, authoring custom AMIs, **Jenkins** pipelines, and **10+** downstream job integrations to expand OS test coverage for endpoint security builds.
- Cut **ARM64** build time ∼**40%** by caching **Bazelisk** in **Docker** volumes on **GitHub Actions** runners.
- Hardened CI **supply-chain security**: removed hard-coded credentials from all **Groovy** Jenkins shared libraries (backward compatible) and added **Rust** cargo-audit scanning.
- Led a **zero-downtime** Bitbucket-to-GitHub migration of **20+ CI pipelines** via mirror clone.

Rocket Mortgage — May 2025 – Sep 2025
Software Engineer Intern, Platform Systems — *Detroit, Michigan*

- Built an **MCP-based AI DevOps agent** in **Go**, using **goroutines** to query GitHub, Azure DevOps, and Slack APIs **concurrently** and surface PR metrics and sprint insights, cutting context-switching by **20 hrs/week**.
- Optimized cross-service data fetching with **Redis** caching and tuned **GraphQL** resolvers, reducing API latency by **35%** under production load for a platform serving **1.5M+ monthly users**.
- Shipped **React/TypeScript** features for Rocket Homes; added **OpenTelemetry** CI/CD metrics.

AssistNow Inc. — May 2024 – Sep 2024
Software Engineer Intern, Backend — *Bentonville, Arkansas*

- Built **Java microservices** on **AWS** (ECS, Lambda, SQS) in the team's migration off a legacy monolith.
- Helped containerize **12+ services** with **Docker** and deploy them on **Kubernetes (EKS)**, cutting environment provisioning from **2 hours to 10 minutes**.
- Built a **C++ gRPC** service with **Protocol Buffers**, cutting cross-service serialization overhead **35%**.

Projects

WatchDog - Real-Time AI Phone Scam Detection | *Python, BERT, LLMs, AWS, Flask* — GitHub | Demo

- **Most Promising Hack** at StarterHacks 2024: a security tool that transcribes live phone calls and warns vulnerable users mid-call when the caller shows scam patterns, reaching **93% precision**.
- Designed a **tiered detection pipeline** (phrase matching → **BERT** classifier → **LLM**) over an ordered **SQS** queue, escalating only ambiguous segments to the costlier models.
- Trained the BERT model on AWS SageMaker (**40%** lower inference latency) and served scoring through event-driven Flask microservices handling **2,000+ requests/min**.

Canify - Rewarding "Made in Canada" | *Python, Flask, Redis, Gemini API, Rust* — GitHub | Demo

- **Top 5 Hack** at Hack Canada 2025 (team of 4): Chrome extension rewarding Canadian-made purchases.
- Built a **Gemini API**-powered product analysis backend (Flask, **Redis**) scoring carts in **<250ms**.

Explainable ECG Arrhythmia Classification | *Python, PyTorch, scikit-learn, SHAP* — GitHub | Report

- Built a reproducible ML pipeline over **100K+ heartbeats**: `pytest`-enforced patient-disjoint split and 3 models.
- Implemented **cost-sensitive Bayes thresholding**, lifting rare-class recall from **0.14 to 0.84**.